LIBRA

Investigating Tritium Breeding on Molten Salts

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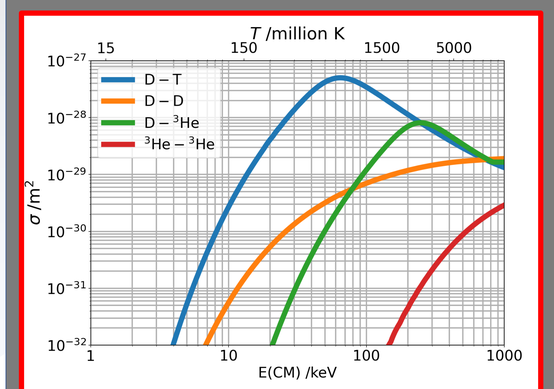
MIT Plasma Science and Fusion Center

Why tritium?

- Deuterium – tritium fusion reaction most promising for energy production:

$$^{2}_{1}D + ^{3}_{1}T \rightarrow ^{4}_{2}He + ^{1}_{0}n + 17.59MeV$$

- Deuterium stable and abundant in nature
- Tritium decays with a half-life of 12.32 years; current global inventory around 20kg
- Tritium is bred as a side product in fission reactors, but production rate is dropping
- To be sustainable, fusion reactors will need to replenish the tritium they use



How do we breed it?

- Neutrons from D-T reaction can react with ^{6}Li, producing tritium:

$$^{6}_{3}Li + ^{1}_{0}n \rightarrow ^{4}_{2}He + ^{3}_{1}T + 4.8MeV$$

- Fusion reactors will be surrounded by ***blankets*** containing ^{6}Li
- Crucial metric – *tritium breeding ratio* (TBR):

$$TBR = \frac{\#\ of\ tritons\ bred}{\#\ of\ tritons\ fused}$$

- Goal of LIBRA: demonstrate TBR>1 at fusion power plant relevant conditions (14MeV neutron energy, 550-700°C, high volumes of molten salt)

Why FLiBe? Why a liquid immersion blanket (LIB)?

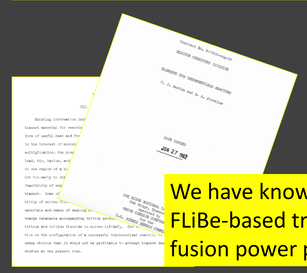
- Favorable neutron interactions on low-Z elements & Be
- FLiBe also functions as the working fluid/coolant
- Small amount of components for large breeder volume
- Easy to drain to allow for maintenance, but no need for FLiBe replacement

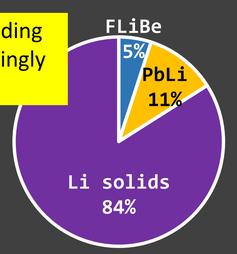
FLiBe LIB enables:

complexity ↓ ↑ temperature

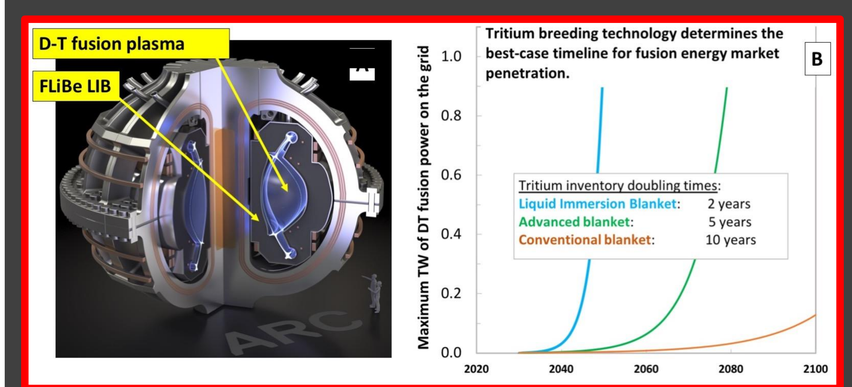
LCOE ↓ ↑ efficiency

Experimental papers on T breeding in the last 50 years overwhelmingly focus on solid Li





We have known about the promise of FLiBe-based tritium breeding for fusion power plants for over 60 years.

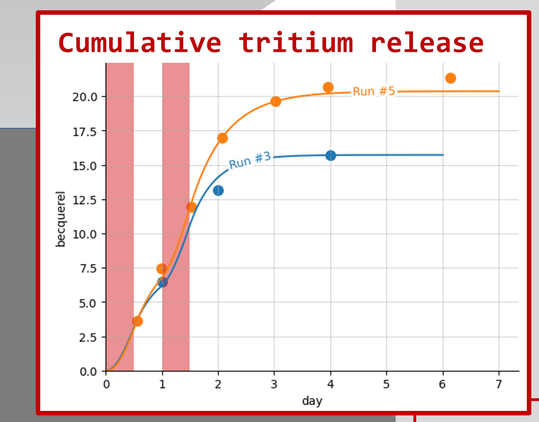


The LIBRA facility

- Requires a new lab space for handling FLiBe capable of supporting neutron shielding
- 4 independent quadrant tanks allow for repeatability and parametric studies
- Short term: demonstrate global TBR measurements
- Long term: study effects of salt chemistry and structural material choices on TBR/T extraction; test novel concepts for T extraction and control

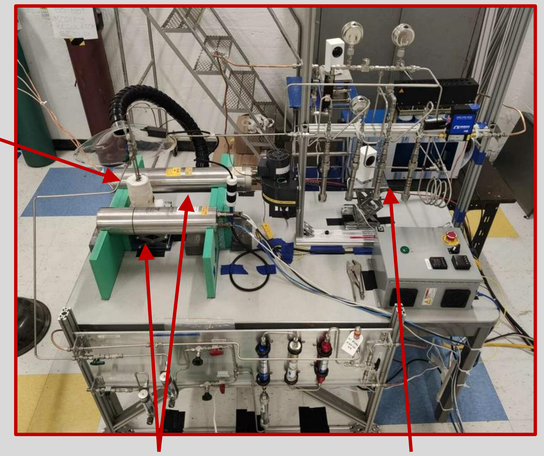
Current experimental progress

- Beryllium toxic, requires high levels of containment
- To avoid hazards, we have used surrogate salts – **ClLiF**
- Neutronics predicts measurable tritium production with 24h irradiation time at 100 ml salt volume



- Salt kept at 700 °C in Inconel 625 crucible
- Two neutron sources of rate ~$3 \times 10^{8}\ n/s$ each
- Dry He gas at low flow rates covering free surface
- Released tritium collected in water bubblers
- Oxidation furnace to distinguish between species
- Samples measured using liquid scintillation counting
- Periodic sampling enables study of release dynamics

We have demonstrated measurable, reproducible tritium production and release from ClLiF in agreement with tritium transport models



- Neutron multiplication on Be in FLiBe necessary for scalability – ClLiF comparable only on small scales
- Next steps:
  - Increase salt volume by an order of magnitude
  - Enable measurement of permeated tritium
  - Build beryllium containment systems
  - Scale up to one quadrant of LIBRA (PIE)

